

Two-way fixed effects estimators with heterogeneous treatment effects*

Clément de Chaisemartin[†] Xavier D’Haultfoeulle[‡]

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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[†]University of California at Santa Barbara, clementdechaisemartin@ucsb.edu

[‡]CREST-ENSAE, xavier.dhaultfoeulle@ensae.fr

This result implies that in general, $\beta_{fe} \neq \delta^{TR}$, so $\hat{\beta}_{fe}$ is a biased estimator of the ATT. To illustrate this, we consider a simple example of a staggered adoption design with two groups and three periods, and where the treatments are non-stochastic: group 1 is untreated at periods 1 and 2 and treated at period 3, while group 2 is untreated at period 1 and treated both at periods 2 and 3.⁸ We also assume that $N_{g,t}/N_{g,t-1}$ does not vary across g : all groups experience the same growth of their number of observations from $t - 1$ to t , a requirement that is for instance satisfied when the data is a balanced panel. Then, one can show that

$$\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot},$$

thus implying that

$$\begin{aligned}\varepsilon_{1,3} &= 1 - 1/3 - 1 + 1/2 = 1/6, \\ \varepsilon_{2,2} &= 1 - 2/3 - 1/2 + 1/2 = 1/3, \\ \varepsilon_{2,3} &= 1 - 2/3 - 1 + 1/2 = -1/6.\end{aligned}$$

The residual is negative in group 2 and period 3, because the regression predicts a treatment probability larger than one in that cell, a classic extrapolation problem with linear regressions. Then, under the common trends assumption, it follows from Theorem 1 and the fact that the treatments are non-stochastic that

$$\beta_{fe} = 1/2E[\Delta_{1,3}] + E[\Delta_{2,2}] - 1/2E[\Delta_{2,3}].$$

β_{fe} is equal to a weighted sum of the ATEs in group 1 at period 3, group 2 at period 2, and group 2 at period 3, the three treated (g, t) cells. However, the weight assigned to each ATE differs from $1/3$, the proportion that each cell accounts for in the population of treated observations. Therefore, β_{fe} is not equal to δ^{TR} . Perhaps more worryingly, not all the weights are positive: the weight assigned to the ATE in group 2 period 3 is strictly negative. Consequently, β_{fe} may be a very misleading measure of the treatment effect. Assume for instance that $E[\Delta_{1,3}] = E[\Delta_{2,2}] = 1$ and $E[\Delta_{2,3}] = 4$. At the period when they start receiving the treatment, both groups experience a modest positive ATE. But this effect builds over time and in period 3, one period after it has started receiving the treatment, group 2 now experiences a large ATE. Then,

$$\beta_{fe} = 1/2 \times 1 + 1 - 1/2 \times 4 = -1/2.$$

β_{fe} is strictly negative, while $E[\Delta_{1,3}]$, $E[\Delta_{2,2}]$, and $E[\Delta_{2,3}]$ are all positive. More generally, the negative weights are an issue if the $E[\Delta_{g,t}]$ s are heterogeneous, across groups or over time.⁹ If $E[\Delta_{1,3}] = E[\Delta_{2,2}] = E[\Delta_{2,3}] = 1$, then $\beta_{fe} = 1 = \delta^{TR}$.

⁸ A similar example appears in Borusyak and Jaravel (2017).

⁹ On the other hand, β_{fe} does not rule out heterogeneous treatment effects within (g, t) cells, as it is identified by variations across (g, t) cells, and does not leverage any within-cell variation.

Here is some intuition as to why one weight is negative in this example. It follows from Equation (7) in the proof of Theorem 1 (see also Theorem 1 in Goodman-Bacon, 2018) that in this simple example, $\beta_{fe} = (\text{DID}_1 + \text{DID}_2)/2$, with

$$\begin{aligned}\text{DID}_1 &= E(Y_{2,2}) - E(Y_{2,1}) - (E(Y_{1,2}) - E(Y_{1,1})), \\ \text{DID}_2 &= E(Y_{1,3}) - E(Y_{1,2}) - (E(Y_{2,3}) - E(Y_{2,2})).\end{aligned}$$

The first DID compares the evolution of the mean outcome from period 1 to 2 in group 2 and in group 1. The second one compares the evolution of the mean outcome from period 2 to 3 in group 1 and in group 2. The control group in the second DID, group 2, is treated both in the pre and in the post period. Therefore, under the common trends assumption, it follows from Lemma 1 in Appendix A (a similar result appears in Lemma 1 of de Chaisemartin (2011) and in Equation (13) of Goodman-Bacon (2018)) that $\text{DID}_1 = E[\Delta_{2,2}]$, but

$$\text{DID}_2 = E[\Delta_{1,3}] - (E[\Delta_{2,3}] - E[\Delta_{2,2}]).$$

DID_2 is equal to the ATE in group 1 period 3, minus the change in group 2's ATE between periods 2 and 3. Intuitively, the mean outcome of groups 1 and 2 may follow different trends from period 2 to 3 either because group 1 becomes treated, or because group 2's ATE changes. The intuition that negative weights arise because $\hat{\beta}_{fe}$ uses treated observations as controls also appears in Borusyak and Jaravel (2017).

We now generalize the previous illustration by characterizing the (g, t) cells whose ATEs are weighted negatively by β_{fe} .

Proposition 1 *Suppose that Assumption 1 holds and for all $t \geq 2$ $N_{g,t}/N_{g,t-1}$ does not vary across g . Then, for all (g, t, t') such that $D_{g,t} = D_{g,t'} = 1$, $D_{.,t} > D_{.,t'}$ implies $w_{g,t} < w_{g,t'}$. Similarly, for all (g, g', t) such that $D_{g,t} = D_{g',t} = 1$, $D_{g,.} > D_{g',.}$ implies $w_{g,t} < w_{g',t}$.*

Proposition 1 shows that β_{fe} is more likely to assign a negative weight to periods where a large fraction of groups are treated, and to groups treated for many periods. Then, negative weights are a concern when treatment effects differ between periods with many versus few treated groups, or between groups treated for many versus few periods.

Proposition 1 has interesting implications in staggered adoption designs, a special case of sharp designs defined as follows.

Assumption 6 (*Staggered adoption designs*) *For all g , $D_{g,t} \geq D_{g,t-1}$ for all $t \geq 2$.*

Assumption 6 is satisfied in applications where groups adopt a treatment at heterogeneous dates (see e.g. Athey and Stern, 2002). In that design, Borusyak and Jaravel (2017) show that β_{fe} is more likely to assign a negative weight to treatment effects at the last periods of the panel.

This result is a special case of Proposition 1: in staggered adoption designs, $D_{g,t}$ is increasing in t , so Proposition 1 implies that $w_{g,t}$ is decreasing in t .¹⁰ Proposition 1 also implies that in that design, groups that adopt the treatment earlier are more likely to receive some negative weights.

Finally, in staggered adoption designs, Athey and Imbens (2018) derive a decomposition of β_{fe} that resembles to, but differs from, that in Theorem 1. They derive their decomposition under the assumption that the dates at which each group starts receiving the treatment are randomly assigned, while we derive ours under a common trends assumption.

3.2 Robustness to heterogeneous treatment effects

Theorem 1 shows that in sharp designs with many groups and periods, $\hat{\beta}_{fe}$ may be a misleading measure of the treatment effect under the standard common trends assumption, if the treatment effect is heterogeneous across groups and time periods. In the corollary below, we propose two robustness measures that can be used to assess how serious that concern is.

Those robustness measures are defined conditional on $\mathbf{D}$, the vector stacking together the treatments of all the (g, t) cells. Specifically, for all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | \mathbf{D})$ denote the ATE in cell (g, t) conditional on $\mathbf{D}$,¹¹ let $\tilde{\Delta}^{TR} = E(\Delta^{TR} | \mathbf{D})$ denote the ATT conditional on $\mathbf{D}$, and let $\tilde{\beta}_{fe} = E(\hat{\beta}_{fe} | \mathbf{D})$. The first measure we consider is the minimal value of the standard deviation of the $\tilde{\Delta}_{g,t}$ s under which one could have that $\tilde{\beta}_{fe}$ is of a different sign than $\tilde{\Delta}^{TR}$. Therefore, this summary measure applies to $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$, rather than β_{fe} and δ^{TR} , the unconditional expectations of $\hat{\beta}_{fe}$ and Δ^{TR} on which we have focused so far. However, one can show that when G , the number of groups, goes to infinity, $\tilde{\beta}_{fe} - \beta_{fe}$ and $\tilde{\Delta}^{TR} - \delta^{TR}$ both converge to 0. So if the number of groups is large, $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$ should not differ much from β_{fe} and δ^{TR} , and our robustness measure “almost” applies to β_{fe} and δ^{TR} .

Let

$$\sigma(\tilde{\Delta}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (\tilde{\Delta}_{g,t} - \tilde{\Delta}^{TR})^2 \right)^{1/2},$$

$$\sigma(\mathbf{w}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1)^2 \right)^{1/2}.$$

¹⁰Borusyak and Jaravel (2017) assume that the treatment effect of cell (g, t) only depends on the number of periods since group g has started receiving the treatment, whereas Proposition 1 does not rely on that assumption.

¹¹ $\tilde{\Delta}_{g,t}$ may differ from $E(\Delta_{g,t})$. To see this, let us consider a simple example where $T = 2$. Then, under Assumption 3, one has $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | D_{g,1}, D_{g,2})$. One may for instance have $E(\Delta_{g,1} | D_{g,1} = 0, D_{g,2} = 0) < E(\Delta_{g,1} | D_{g,1} = 1, D_{g,2} = 1)$, if a group is more likely to be treated if her treatment effect is initially high.